Robert Alan Clements

Assistant Professor at University of San Francisco
San Francisco Bay Area, CA USA
rclements@usfca.edu
r-clements linkedin.com/in/rbclements/ | *Updated:* July 18, 2026

EDUCATION

University of California, Los Angeles, Ph.D. Statistics 2011

- Thesis Topic: *A Comparison of Residual Analysis Methods for Space-time Point Processes with Applications to Earthquake Forecast Models*
- Advisor: Professor Frederic Paik Schoenberg
- Area of Study: Point processes

University of California, Los Angeles, M.S. Statistics 2009
Humboldt State University, B.A. Mathematics 2006

- Cum Laude
- Emphasis: Applied Mathematics

FELLOWSHIPS

Eugene V. Cota-Robles Fellowship 2007–2011
Role: Graduate Fellow

EMPLOYMENT

University of San Francisco, San Francisco, CA

Assistant Professor, MS in Data Science Aug 2022–Present

Director of the Center for AI and Data Ethics Aug 2023–Present

Practicum Director Aug 2022–July 2024

Optum, San Francisco Bay Area, CA

Senior Director of Data Science Aug 2021–July 2022

Director of Data Science Mar 2019–Jul 2021

University of California, Berkeley Extension, San Francisco office

Instructor Jan 2018–Jan 2020

Walmart Labs, San Bruno, CA

Data Scientist - Data Science Manager Aug 2018–Mar 2019

UnitedHealthcare, San Francisco Bay Area, CA

Senior Principal Data Scientist Aug 2016–Aug 2018

GE Digital, San Ramon, CA

Staff Data Scientist Mar 2016–Jul 2016

Senior Data Scientist — Dec 2014–Mar 2016

Verisk Analytics, San Francisco, CA

Senior Statistician — Sep 2013–Nov 2014

GFZ Potsdam, Potsdam, Germany

Postdoctoral Researcher — Sep 2011–Jun 2013

TEACHING

Master of Science in Data Science, University of San Francisco

Primary Instructor:

Introduction to Machine Learning

Probability and Statistics

Machine Learning Operations

Data Ethics

Communications for Data Science

UC Berkeley Extension

Primary Instructor:

Introduction to Data Science

PUBLICATIONS: PEER REVIEWED

Clements RA, Schoenberg FP, and Schorlemmer D. 2011. Residual analysis methods for space-time point processes with applications to earthquake forecast models in California. *Annals of Applied Statistics*. 5 (4): 2549-2571. doi:10.1214/11-AOAS487

Clements RA, Schoenberg FP, and Veen A. 2012. Evalution of space-time point process models using super-thinning. *Environmetrics*. 23: 606-616. doi:10.1002/env.2168

Holschneider M, Zöller G, **Clements RA**, and Schorlemmer D. 2014. Can we test for the maximum possible earthquake magnitude? *Journal of Geophysical Research: Solid Earth*. 119(3), 2019-2028. doi:10.1002/2013JB010319

Schneider M, **Clements RA**, Schorlemmer D, and Rhoades D. 2014. Likelihood- and Residual-Based Evaluation of Medium-Term Earthquake Forecast Models for California. *Geophysical Journal International*. 198 (3): 1307-1318. doi:10.1093/gji/ggu178

Mak S, **Clements RA**, and Schorlemmer D. 2014. The Statistical Power of Testing Probabilistic Seismic-Hazard Assessments. *Seismological Research Letters*. v. 85 no. 4 p. 781-783. doi:10.1785/0220140012

Mak S, **Clements RA**, and Schorlemmer D. 2014. Comment on "A New Procedure for Selecting and Ranking Ground-Motion Prediction Equations (GMPEs): The Euclidean Distance-Based Ranking (EDR) Method" by Ozkan Kale and Sinan Akkar. *Seismological Research Letters*. 104 (6): 3139. doi:10.1785/0120140106

Gordon SJ, **Clements RA**, Schoenberg FP, Schorlemmer D. 2015. Voronoi residuals and other residual analyses applied to CSEP earthquake forecasts. *Spatial Statistics*. 14B: 133-150. doi: 10.1016/j.spasta.2015.06.001

Mak S, **Clements RA**, and Schorlemmer D. 2015. Validating Intensity Prediction Equations for Italy by Observations. *Bulletin of the Seismological Society of America*. 105(6): 2942-2954. doi:10.1785/0120150070

Mak S, **Clements RA**, and Schorlemmer D. 2017. Empirical Evaluation of Hierarchical Ground-Motion Models: Score Uncertainty and Model Weighting. *Bulletin of the Seismological Society of America*. 107(2): 949-965. doi:10.1785/0120160232

Clements RA and Thiebaut N. 2026. But Have You Ever Deployed a Model to Production? Experiences with Teaching Machine Learning Operations in a Data Science Curriculum. In *2026 IEEE/ACM 48th International Conference on Software Engineering (ICSE-SEET '26), April 12–18, 2026, Rio de Janeiro, Brazil*. ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3786580.3786950

Clements RA, Makam, SM, Dixon, H, Shen Y. 2026. Ethical guidelines for machine learning competitions. *AI Ethics*. 6, 387. https://doi.org/10.1007/s43681-026-01201-4

PUBLICATIONS: UNDER REVIEW

Clements RA. deployml: Supporting MLOps Education Through Automated Infrastructure Deployment.

Clements RA, Schoenberg F. Adaptive Voronoi Residual Diagnostics for Spatiotemporal Point Processes.

Ahrweiler P, Barakova E, Clements RA, Crockett K, Dhurandhar A, Dogan I, Dollinger M, Farahi A, Ferguson S, Gunes H, Hajiramezanali E, Kazeminasab S, Parrish A, Perez Ortiz M, Varshney K. Beyond Explainable AI (XAI): Paradigm Shift and Post-XAI Research Paths

PRESENTATIONS (ORAL AND POSTER)

Evaluation of Earthquake Prediction Models Using Residual Analysis for Spatial Point Processes (Contributed oral), 2009 Joint Statistical Meetings, August 2009, Washington DC.

Model Assessment for Space-Time Point Processes Using Super-Thinning (Contributed oral), 2010 Joint Statistical Meetings, August 2010, Vancouver, Canada.

Residual Analysis of Space-time Point Processes with Applications to Earthquake Forecast Models in California (Poster), 7th International Workshop on Statistical Seismology, May 2011, Santorini, Greece.

The Testability of Mmax Estimates (Poster), European Geosciences Union General Assembly 2012, April 2012, Vienna, Austria.

Yeh, C. J., Clements, R., Nolasco, E., Pubill, C., & Bautista, L. (2024). Using Community Cultural Wealth to Promote College Access Program among BIPOC High School Students, Poster presented at the American Psychological Association Convention, Seattle, WA.

Yeh, C. J., Murrieta, V., & Clements, R. (2024). Dismantling Barriers to Education: The Role of School Counselors in Advancing Equity through Cultural Capital, Conference proposal submitted to 14th International Conference on Education and Justice, Oahu, HI.

Can You Explain Yourself? Transparency and Explainable AI (Invited oral), NRT Ethical AI Talk Series at UT Austin, November 2024, Remote

WORKSHOPS

An Ethics Framework for Journalism and Generative AI (Invited), World Press Institute and USF Data Institute AI Ethics Workshop, November 2024, San Francisco, CA.

UNIVERSITY AND DEPARTMENT SERVICE

Practicum Mentor for 45+ students, 2022-present.

Committee Member, Term Faculty Search, MS in Data Science Program, 2024.

Affiliated Faculty with the Center for Law, Tech, and Social Good, 2024-present.

Faculty Director of the Center for AI and Data Ethics, 2023-present.

Faculty Practicum Director, 2022-2024.

Podcast host for the USF Data Science Podcast, 2022-present.

Committee Member, Tenure-Track Faculty Search, MS in Data Science Program, 2022.

SOFTWARE

deployml-core: a python library for deploying end-to-end ML pipelines in GCP

stppResid: an R package for residual analysis of space-time point process models, 2013

PATENTS

US Patent 20180307784, "Usage Based Lifing" Oct 2018
Inventors: Stevens, Craig Wesley (Revere, MA, US), Chan, Simon Shu Kong (Danville, CA, US), Wu, Siyu (San Ramon, CA, US), Vahldick, Lauren Ashley (Salem, MA, US), Wight, Ronald Burton (Lisbon Falls, ME, US), **Clements, Robert Alan (Pacifica, CA, US)**